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Effects of the Reynolds number and structural damping on vortex-induced vibrations of elastically-mounted rigid cylinder

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ABSTRACT

In this work, analyses of the damping ratio and turbulence effects on three-dimensional vortex-induced vibrations of a two degrees-of-freedom elastically mounted rigid cylinder are carried out. Simulations are conducted for damping ratios of $\zeta = 0.5$, 2.5, 5.0 and 10.0% for flows covering Reynolds numbers in the three TrSL flow regimes, in the range of $Re = 7.5 \times 10^2$ to 1.3×10^5. Variation of the response amplitudes, phase angles, force coefficients and frequency ratios by the damping ratios and Reynolds number are investigated using the classically damped spring-mass system equations programmed inside the CFD software Fluent. New results are obtained for very-high Reynolds numbers and different damping ratios. New conclusions are drawn from the mechanical responses using surface interpolation of 240 data-points, which allows the investigation of the trends of a given response variable ξ based on the A_y^* *vs.* $\zeta(m^* + C_A)$ *vs.* ξ three-dimensional plot proposed by the authors. A new mechanism of elongation of the upper branch extension based on the universal reduced damping, S_u, during the TrSL3 regime is described.

1. Introduction

Vortex-induced vibrations (VIV) of circular cylinders is an important subject for offshore and aerospace applications, since sub-sea cables, pipelines and risers are subjected to intense vibration loads due to vortices formation, and the present study builds on our effort to characterize mechanical responses of cylinders undergoing VIV in moderate and high Reynolds regimes in order to improve our understading of this phenomenon using more realistic turbulence conditions. Pionnering works, such as those of Feng [1] and Khalak and Williamson [2], have investigated this phenomenon concerning the influences of non-dimensional groups on the response phase and amplitude of the cylinder. They have concluded that this phenomenon is mainly governed by the damping-mass parameter, defined as $m^*\zeta$, where ζ is the structural damping and m^* is the mass ratio, given by $m^* = 4m/\rho_f \pi D^2$, in which ρ_f is the specific mass of the fluid and D, the diameter of the cylinder.

If an elastically mounted cylinder is subjected to an external flow, it experiences fluctuating lift and drag forces caused by asymmetric vortex formation, which may synchronize with its natural frequency, leading to a resonance state called by "lock-in". Further, it was demonstrated that the lock-in occurs over a specific regime of reduced velocities, V^*, defined as $V^* = V_\infty / f_{n,w} D$, where V_∞ is the free-stream velocity and $f_{n,w}$, the natural frequency of the cylinder in water.

For 1DOF systems with high damping-mass parameters, investigated by Feng [1], who have worked with $m^*\zeta = 36.0\%$, two response branches for the A_y^* *vs.* V^* plot are found, namely the "initial" and "lower" branches, as it is shown in Fig. 1. A_y^* is defined as the ratio between the cross-flow movement amplitude and the cylinder diameter, $A_y^* \equiv A_y/D$. Experiments conducted by Khalak and Williamson [2] with a cylinder of low damping-mass value, $m^*\zeta = 1.08\%$, showed three distinct branches: the initial, upper and lower branches, which are also shown in Fig. 1. The new intermediate state, the upper branch, presents high reduced amplitudes of up to $A_y^* = 1$. In addition, it is possible to conclude from the superposition of both A_y^* curves that the synchronization regime is incremented by the decrease in the mass ratio value (see also [3] and [4]).

In a major advance, Jauvits and Williamson [5] experimentally investigated the 2DOF elastically mounted circular cylinder allowed to move in the cross-flow, y, and stream-wise, x, directions. In their study, they were able to observe response amplitudes of up to $A_y^* = 1.5$, associated with the formation of what they called by super-upper branch. It was observed that the x-displacement had small amplitudes of about $A_x^* = 0.3$. In addition, the peak response amplitude shifts from $V^* = 6$ to $V^* = 8$ when the second DOF is allowed to move.

The mechanical responses described above are intrinsically associated with the formation and shedding of the wake structures.

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List of symbols

A_i^*	ith DOF reduced amplitude A_i/D
α	Modified damping-mass parameter $\zeta(C_A + m^*)$
b_i	ith DOF damping coefficient
C_A	Added mass coefficient
C_i	ith DOF force coefficient $2F_i/\rho_f V^2 DL$
D	Cylinder diameter
F_i	ith DOF excitation force
f_n	Natural frequency in vacuum
$f_{n,w}$	Natural frequency in water
k_i	ith DOF equivalent spring stiffness
L	Cylinder's length
m	Cylinder's mass
μ	Dynamic viscosity
m^*	Mass ratio $4m/\rho_f \pi D^2 L$
$m^*\zeta$	Damping-mass parameter $4m\zeta/\rho_f \pi D^2 L$
ϕ_i	Phase difference between ith DOF excitation force and displacement
r	Frequency-ratio f_x/f_y
r_n	Natural frequency-ratio $f_{n,x}/f_{n,y}$
r_i	ith DOF frequency-ratio $f_i/f_{n,i}$
Re	Reynolds number $\rho_f V_\infty D/\mu$
ρ_f	Fluid density
S_u	universal reduced damping $\pi^3\zeta/V^{*2}$
θ	Phase difference between x and y displacements
V_∞	Free-stream velocity
V^*	Reduced velocity $V_\infty/f_{n,w}D$
ζ	Damping ratio $b_i/2\sqrt{k_i m}$

Experiments conducted by Williamson and Roshko [6] and by Morse and Williamson [7] showed that the initial-excitation regime, represented in Fig. 1, is associated with the 2S vortex shedding mode. During the shot up to the upper branch and along the lower branch extension, the system presents the 2P mode. Sequentially, the sudden decrease observed in the response amplitude is due to the system reaching the region with no synchronized vortex shedding patterns, occurring the desynchronization between the excitation forces and the responses of the cylinder. In the literature, 2S stands for two single vortices being shed from the cylinder by semi-cycle of oscillation and 2P, for a pair of vortices being shed parallel from its surface. Further, the 2DOF system investigated by Jauvits and Williamson [5] presented a 2T wake structure, standing for three vortices being shed from the surface of the cylinder by semi-cycle of oscillation.

The results from the studies cited above, among others (see [3,4,8]), have created a solid foundation for the development of new engineering applications that relate to the effects of vortex-induced vibration on circular cylinders. The state-of-art of modern studies are directed towards investigating response changes due to the Reynolds number [9,10], three-dimensional effects in the fluid-structure interaction [11] and modification in the wake structures by the ratio between the cross-flow and the stream-wise natural frequencies, $r_n = f_{n,y}/f_{n,x}$ [12]. Other topics of interest to the scientific community are vibration control [13], energy extraction from VIV [13,14] and the effects caused by the presence of wall boundaries close to the cylinder [15–19]. Most of published studies on VIV of cylinders are based on experimental studies [20–22]. Some numerical results based on two-dimensional computational fluid-dynamics (CFD) have also started to be carried out in the last fifteen years for low and medium Reynolds numbers [23–25]. For three-dimensional CFD models, a growing number of studies have been conducted in recent years, as well [26–28]. The present work implements three-dimensional CFD simulations to investigate VIV of elastically mounted rigid cylinders with 2DOF subjected to low, medium and high Reynolds numbers flows.

Zdravkovich [29] subdivided the flow around smooth cylinders into 15 flow regimes based on the characteristics of the boundary layers and the separated shear layer. Some of these regimes, classified by Reynolds number intervals, are listed in Table 1. Key features of each regime are also given. Vortex-induced vibrations have not been entirely studied in the TrSL3 regime and there is lack of numeric and experimental data for flows in the TrSL2 and TrSL3 regimes, in which the acronym TrSL stands for the "transition of shear layer". The present study covers the TrSL1, TrSL2 and TrSL3 flow regimes and, hereafter, the "low", "medium" and "high" Reynolds flows refer respectively to these regimes. Remarkable differences are observed in the fluid-structure interaction phenomena in the different Reynolds regimes described, mainly along the near-wake domain, in which the interactions between the cylinder and the flow occur (see Roshko [30]). The results obtained in the present study are compared to those experimentally obtained by some of the studies listed in Table 2, in which the reduced amplitude is the basis of the comparative study.

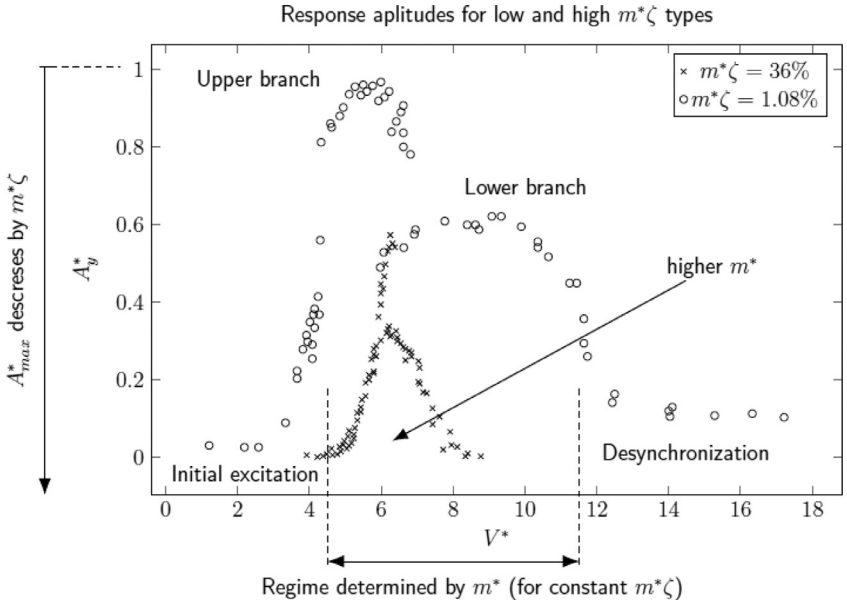


Fig. 1. Classic 1DOF response amplitudes obtained by (**o**) Khalak and Williamson [2] with a low damping-mass parameter of $m^*\zeta = 1.08\%$, and (**x**) by Feng [1], who have worked with a high damping-mass parameter of $m^*\zeta = 36\%$.

Table 1
Classification of flow regimes based on the wake structures of circular cylinders. Table adapted from Raghavan and Bernitsas [31].

Reynolds number	Regime	Feature
Laminar regime		
$Re < 1$		Creeping flow
$3-5 < Re < 30-40$	L	Steady separation
$30-40 < Re < 150-300$	L	Periodic laminar shedding
Sub-critical regime		
$150-300 < Re < 200-250$	TrW1	Transition of laminar vortices in wake
$200-250 < Re < 350-500$	TrW2	Transition of irregular vortices during formation
$350-500 < Re < 1-2\times10^3$	TrSL1	Development of transition waves in free shear layer
$1-2\times10^3 < Re < 1-2\times10^4$	TrSL2	Formation of vortices in free shear layer
$1-2\times10^4 < Re < 1-2\times10^5$	TrSL3	Fully turbulent shear layer
Critical regime		
$1-2\times10^5 < Re < 3.1\times10^5$	TrS0/TrBL0	Onset of transition at separation point
$3.1\times10^5 < Re < 3.4\times10^5$	TrS1/TrBL1	Single separation bubble regime
$3.4\times10^5 < Re < 3.8\times10^5$		Unstable regime

Table 2
Data from experimental studies of vortex-induced vibration in the literature.

Researcher	Reynolds	Regime	$m^*\zeta$(%)	$A^*_{y,\max}$
Klamo et al. [34]	5.2×10^2	TrSL1	0.039	0.55
Gonçalves et al. [35]	$1.4\times10^3-5.0\times10^4$	TrSL2	0.262	1.50
Jauvits and Williamson [5]	$7.2\times10^3-1.5\times10^4$	TrSL2	0.640	1.50
Kim et al. [36]	$4.8\times10^3-2.8\times10^4$	TrSL2	4.290	1.70
Raghavan and Bernitsas [31]	$8.0\times10^3-1.5\times10^5$	TrSL2/TrSL3	0.251	1.97
Present study	$7.5\times10^2-1.3\times10^5$	TrSL1/TrSL2/TrSL3	1.200	1.70

The TrSL1 flow regime, which has been the focus of many studies based on two-dimensional computational fluid-dynamics, is characterized by the lowest drag force coefficients of the entire TrSL regime due to the elongation of the vortex formation region and the development of transition waves in free shear layers. In the TrSL2 flow regime, the transition in the shear layer from laminar to turbulent is initiated and completed and the formation of transition vortices in free shear layers shortens the vortex formation region, increasing the drag coefficient. Further, as it was observed by Raghavan and Bernitsas [31], the Strouhal number is approximately constant and equal to 0.2 in the TrSL2. In the TrSL3, the shear layers are fully turbulent. This last stage of the transition from laminar to turbulent flows in the shear layers is characterized by the invariance of the drag and lift coefficients, C_x and C_y, due to the formation of a relatively short and invariable vortex formation region. Also, the lowest values of the drag coefficient are oberved in the TrSL3 regime. These flow regimes are further investigated by Unal and Rockwell [32] and Zdravkovich [33], in which excellent conceptual overviews and summaries of turbulent flows past circular cylinders are provided.

Further, as described by Wei and Smith [37] and Williamson [38], flow properties other than the pressure coefficients are well characterized in the laminar, sub-critical and critical Re regimes, which are listed in Table 2. During laminar flows, viscous forces are dominant and the flow is characterized by smooth and constant motion. Sub-critical flows are characterized by laminar separation transition in shear layer. It should also be mentioned that the TrSL regimes are three out of five sub-divisions of the sub-critical flow regime. In the critical regime, turbulent separation of wake structures are observed. Further, one of the motivations of the present study is related to the lack of experimental and numeric data for VIV in high Reynolds numbers flows, mostly for $Re > 10^4$, in the end of the sub-critical regime. The analyses of flows in the given conditions may create new paradigm for future studies concerning the effects of the Reynolds number on VIV of elastically mounted circular cylinders.

In the present work, three-dimensional CFD simulations are grouped by Reynolds number regimes and damping ratio values. Four sets of Reynolds number intervals covering $7.5\times10^2-3.3\times10^3$, $2.5\times10^3-1.2\times10^4$, $7.5\times10^3-3.5\times10^4$ and $2.5\times10^4-1.3\times10^5$ are investigated using $\zeta = 0.5$, 2.5, 5.0 and 10.0% each. Simulations are conducted for reduced velocities in the range of $V^* = 3-15$. The structural damping coefficient, mass and stiffness values are added in the classically damped spring-mass system within the ANSYS solver. Low to moderate damping-mass values of $m^*\zeta = 1.2$, 6.0, 12.0 and 24.0% are tested. Results are obtained using a computationally efficient structured mesh with a new approach proposed to perform mesh deformation, which will be detailed in Section 2.3. Numeric results are validated using experimental data from the studies listed in Table 2. Simulations are carried out for computational times of about 24h using a desktop IBM PC with six cores at 1.9 GHz. New sets of data are obtained for medium and high Re numbers, considering three-dimensional flows around circular cylinders with 2DOF. Responses of the cylinder are analysed based on their behaviour in the $\zeta(C_A + m^*)$ $vs.$ V^* plane, which has not been performed by other studies. The modified damping-mass parameter, $\zeta(C_A + m^*)$, is implemented to reduce the amplitude of motion to its non-dimensional form, A_i^*, which makes it an important parameter for understating VIV of cylinders (see [2]).

The present paper is organized as follows: the second section covers numeric method, structural model, computational domain and mesh sizing. The third section covers results for displacements, forces and phase angles, investigating the VIV phenomenon for a wide range of Reynolds numbers. A subsection is designed to investigate the near-wake regime.

2. Numerical method

2.1. Fluid model

The fluid flow is governed by a RANS (Reynolds averaged Navier–Stokes equations) model [39]. Solution variables in the exact Navier–Stokes equations are decomposed into the mean time-averaged and fluctuating components, such as $\eta = \overline{\eta} + \eta'$, and are substituied into the instantaneous continuity and momentum equations, which are re-written as [40]:

$$\frac{\partial \overline{u}_i}{\partial x_i} = 0 \tag{1}$$

$$\frac{\partial \overline{u}_i}{\partial t} + \frac{\partial}{\partial x_j}\left(\overline{u}_i\overline{u}_j + \overline{u'_i u'_j}\right) = -\frac{1}{\rho_f}\frac{\partial \overline{p}}{\partial x_i} + \frac{\partial \overline{\tau}_{ij}}{\partial x_j} \tag{2}$$

considering an incompressible flow. In Eq. (2), ρ_f and p are the fluid density and pressure, respectively. The subscripts i and j indicate the ith and jth components of the Cartesian coordinates. Also, τ_{ij} is the viscous stress tensor. The numerical solution algorithm is a Pressure-Based type. Fluid properties are estimated first, then the pressure-correction equation is obtained from the continuity equation (Eq. (1)). Mass flux, pressure and velocity are iteratively updated before continuing to solve turbulence and other scalar quantities. These steps are made in loop until the convergence criteria are achieved. The governing equations are discretized using finite volumes. The method for integrating equations is the SIMPLE. A second order backward Euler scheme was adapted for the temporal discretization of pressure, velocity and momentum equations [41].

The relative convergence criteria for all variables, including the displacements of the cylinder, considers a tolerance of 2.5×10^{-4}. The turbulence model is the Standard SST $k-\omega$, which stands for Shear-Stress-Transport. This $k-\omega$ variant makes the numerical method more robust for dealing with flow separation under adverse pressure gradients by including the transport effects into Eq. (2) [42]. In order to reduce the tendency of over-predict the turbulence production, the Kato-Launder and production limiter modifications were introduced into the solver. The Kato-Launder modification reduces the tendency that most two-equations models have to over-predict the turbulent production in large normal strain flows, which is a well-known issue on VIV of cylinders. The same reason justifies the use of the production limiter modification (please read [43] and [44]).

2.2. Structural model

For an elastically mounted circular cylinder, the motion at the ith DOF is described by:

$$m\ddot{x}_i(t) + b_i\dot{x}_i(t) + k_i x_i(t) = F_i(t) \tag{3}$$

where m, b_i and k_i are the associated mass, structural damping and stiffness. The hydrodynamic forces, $F_i(t)$, are obtained by the CFD solver. Sequentially, Eq. (3) is integrated twice for the ith DOF to obtain the velocity, $v_{i,t+1}$, and displacement, $x_{i,t+1}$, of the center of gravity of the cylinder in the time step $t+1$. The following equations result from the integration process of Eq. (3):

$$\begin{aligned}
\ddot{x}_{i,t+1} &= \tfrac{1}{m}\left(F_{i,t} - b_i\dot{x}_{i,t} - k_i x_{i,t}\right)\\
\dot{x}_{i,t+1} &= \dot{x}_{i,t} + \tfrac{\Delta t}{24}(9\ddot{x}_{i,t+1} + 19\ddot{x}_{i,t} - 5\ddot{x}_{i,t-1} + \ddot{x}_{i,t-2})\\
x_{i,t+1} &= x_{i,t} + \tfrac{\Delta t}{24}(9\dot{x}_{i,t+1} + 19\dot{x}_{i,t} - 5\dot{x}_{i,t-1} + \dot{x}_{i,t-2})
\end{aligned} \tag{4}$$

in which rates are determined using a fourth-order multi-point Adams-Moulton formulation (see [41]).

If the forcing is assumed to be sinusoidal in nature, $F_i(t) = F sin(\omega t + \phi_i)$, where ϕ_i is the phase angle between the ith DOF displacement and excitation force and ω is the frequency of vibration, it may be subdivided into two components: $F_i(t) = -L_a\ddot{x}_i(t) + L_v\dot{x}_i(t)$, where the former represents the forces in phase with the acceleration, and the latter, in phase with the velocity of the body. L_a and L_v are coefficients that depend on the reduced amplitude, A_i^*, and ω. Eq. (3) may be re-written as:

$$m\ddot{x}_i(t) + b_i\dot{x}_i(t) + k_i x_i(t) = -\frac{\pi}{4}\rho_f D^2 C_{a,i}\ddot{x}_i(t) + \frac{1}{2}\rho_f D V_\infty^2 C_{v,i}\dot{x}_i(t) \tag{5}$$

where $C_{v,\,i}$ and $C_{a,i}$ are the coefficients of the hydrodynamic forces in phase with the velocity and acceleration of the body, respectively. Hereafter, the convention $C_{a,i} \equiv C_A$ is adopted for the added mass coefficient, which is also considered as ideal, $C_A = 1$. Further, we work with a system that has the same natural frequency in water in both stream-wise and cross-flow directions, which is determined by:

$$f_{n,w} = \frac{1}{2\pi}\sqrt{\frac{k}{m + \rho_f D^2 \pi C_A/4}} \tag{6}$$

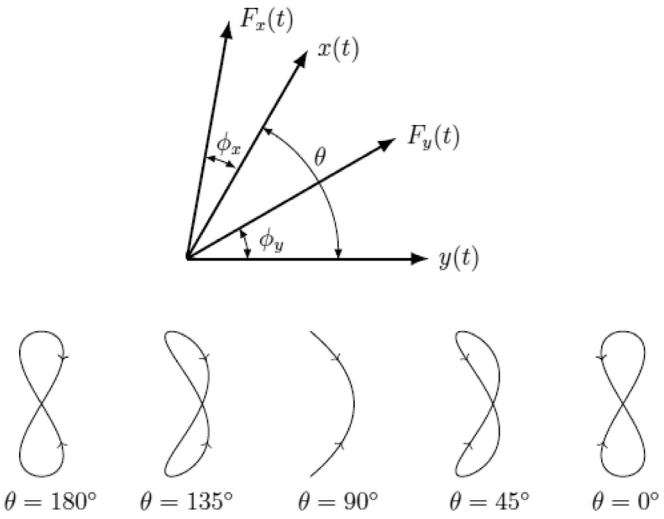


Fig. 2. Typical xy-trajectories as function of the phase θ between x and y displacements. ϕ_x and ϕ_y phase angles are defined as well. Illustrations were made using $x(t) = \sin(2\omega t + \theta)$ and $y(t) = 0.3\sin(\omega t)$.

In the present study, we shall find correlations between wake patterns and XY motion. Besides this, it is focused on how does the coupling between hydrodynamic forces and the dynamics of the cylinder happens for different structural damping values under diverse flow regimes. Furthermore, as noticed by Khalak and Williamson [4], the overall response for this 2DOF model, given by Eq. (3), is well represented by the equations:

$$x(t) = X\sin(2\omega t + \theta) \quad \text{and} \quad y(t) = Y\sin(\omega t) \tag{7}$$

where X and Y are the vibration amplitude of the cylinder in the stream-wise and cross-flow direction, respectively, and θ is the phase between x and y displacements. The phase angle between the displacement, x, and excitation forces parallel to it, F_x, is given by:

$$\phi_x = arccos\left(\frac{\langle x, F_x\rangle}{\|x\| \cdot \|F_x\|}\right) \tag{8}$$

which can be similarly obtained for ϕ_y. Fig. 2 shows the overall response of an elastically mounted 2DOF cylinder subjected to external flow for different θ angles.

By substituting Eq. (7) into Eq. (5), it is possible to obtain non-dimensional forms for the response amplitudes, A_i^*, given by [5]:

$$A_y^* = \frac{C_y\sin(\phi_y)}{4\pi^3(m^* + C_A)\zeta}\left(\frac{V^*}{r_y}\right)^2 r_y, \quad A_x^* = \frac{C_x\sin(\phi_x)}{4\pi^3(m^* + C_A)\zeta}\left(\frac{V^*}{2r_y}\right)^2 2r_y \tag{9}$$

where C_i is the force coefficient in the ith direction, given by $C_i = 2F_i/\rho_f L D V_\infty^2$, and r_i is the oscillation frequency ratio in the ith direction, $r_i = f_i/f_{n,i}$, in which f_i is the vibration frequency and $f_{n,i}$ is the natural vibration frequency associated with the ith DOF. In this paper, it is focused on capturing the different response amplitudes of elastically mounted rigid cylinders undergoing external flows on diverse Reynolds regimes. Other mechanical responses, such as phase angles, vibration frequencies, among others, are also investigated in the $\zeta(m^* + C_A)$ $vs.$ V^* plane, isolating and characterizing the effects of structural damping and turbulence on VIV of elastically mounted rigid cylinders, which brings a new perspective on VIV that has not yet been investigated by the literature.

2.3. Computational mesh and time-step

The computational domain was scaled based on geometric characteristics verified by several studies [45–48]. The ratio between the length of the cylinder and its diameter was set to $L/D = \pi$, which is sufficient to obtain accurate hydrodynamic forces and displacements

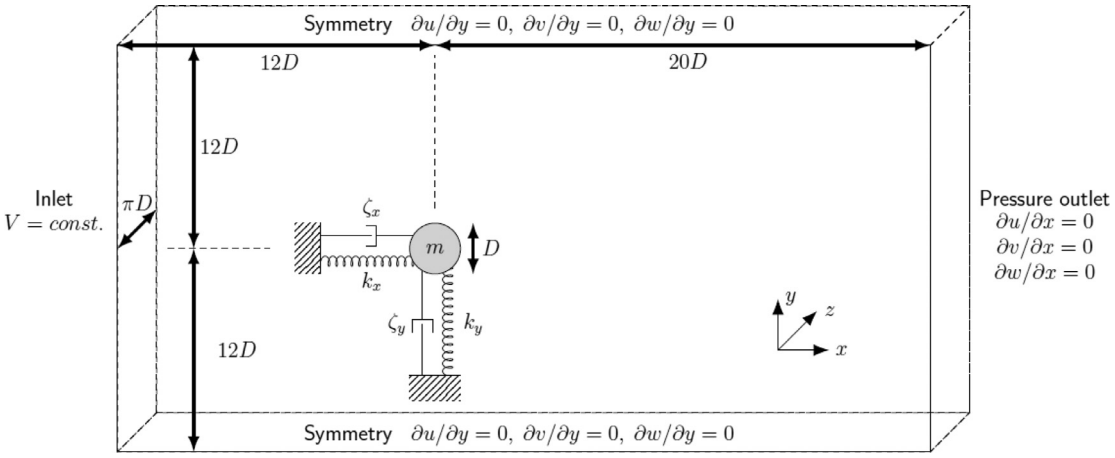


Fig. 3. Characteristics of the fluid domain. The components of the velocity field in the *x, y* and *z* directions are given by *u, v* and *w*, respectively.

Table 3
Results of the mesh analysis for $Re = 1.3 \times 10^5$ and $V^* = 15$.

Mesh	Number of elements	n_z	f_y/f_n	A_y^*	Time (h)
Coarse	3.6×10^5	20	3.018	1.685	24
Medium	4.6×10^5	30	3.128(+3.5%)	1.685(+0%)	39(+62.5%)
Dense	9.3×10^5	60	3.100(−0.9%)	1.668(−1.0%)	228(+484.6%)

(see Batcho et al. [49]). The velocity-inlet and pressure-outlet, illustrated in Fig. 3, have been positioned far from the cylinder to avoid influences of the boundaries on the near-wake and vortex-street regions. The symmetry boundary condition has been implemented in the upper, lateral and lower faces of the domain. The total domain size is $32D \times 24D \times \pi D$ in the stream-wise, cross-flow and transverse directions, respectively. The cylinder is $12D$ down-stream the velocity-inlet

The number of elements in the circumferential direction of the cylinder was set to 140, based on the study conducted by Zhao et al. [48]. The height of the first cells adjacent to the cylinder surface, Δy, was selected based on the $y^+ = 1$ criterion, since the Standard $SST\ k-\omega$ turbulence model is implemented. The parameter was estimated using $y^+ = \Delta y(u_t/\nu)$, where u_t is the friction velocity, given by $u_t = \sqrt{\tau_w/\rho_f}$, and ν is the dynamic viscosity. τ_w is the local shear stress. Since three different regimes of Reynolds numbers are covered in this study, the height of the first cells adjacent to the cylinder wall were selected using the expression $\Delta y = 2.5 \times 10^{-n}$, where $n = 3$, for the low, $n = 4$, for the medium and $n = 5$, for the high Reynolds case. The mean value of y^+ was found to be $\overline{y}^+ = 0.15$ for the wall of the cylinder, for the highest Reynolds number in each set.

Three mesh sizes were tested for this work. Since the dimensioning of the mesh parallel to the *xy* plane (see Fig. 3) is based on the y^+ criterion, the only free variable was the number of elements in the transverse direction, defined as n_z. The three meshes tested have $n_z = 20$, for the "coarse", $n_z = 30$, for the "medium" and $n_z = 60$, for the "dense" structure, which selected characteristics are listed in Table 3. Based on both cross-flow vibration frequency and movement amplitude changes produced by the refinement of the mesh, the "medium" structure is considered appropriate to evaluate the VIV phenomenon, which also provides appreciable low computational time. The time-step was selected as a small fraction of the period of free vibration of the cylinder in vacuum. According to the study conducted by Gsell et al. [26], the value of $\Delta t = (1/f_n)/200$ is considered sufficiently small. Simulations were programmed to reproduce 20 periods of vibrations, requiring 4000 time-steps each.

The fluid domain, represented in Fig. 3, is subdivided into two regions, as it is shown in Fig. 4. The outer partition, with thickness of one element height, absorbs all movement of the inner in form of deformation, based on an iterative procedure to update the nodal positions using equivalent spring force equations. The inner partition moves as a rigid body accordingly to the displacements of the cylinder, which dynamics is determined by Eq. (3). This approach allowed us to reduce the computational times to values comparable to those of two-dimensional systems (see [50] and [51]). In addition, a large portion of the domain is

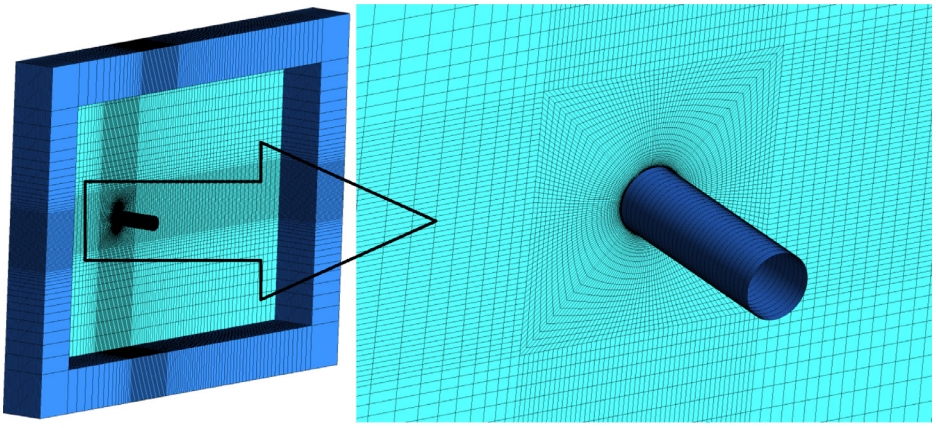


Fig. 4. Details of the computational mesh developed for this study. The intern region moves as a rigid body accordingly to the cylinder. The external region absorbs the movement in form of deformation.

Table 4
Numerical dataset.

Case	Nomenclature	Reynolds range	Regime	ζ	$m^*\zeta$	f_n
1-1	Low Re	$7.5\times10^2-3.3\times10^3$	TrSL1/TrSL2	0.5%	1.2%	0.1 Hz
1–2	Medium Re	$2.5\times10^3-1.2\times10^4$	TrSL2	0.5%	1.2%	0.3 Hz
1–3	Medium-high Re	$7.5\times10^3-3.5\times10^4$	TrSL2/TrSL3	0.5%	1.2%	1.0 Hz
1–4	High Re	$2.5\times10^4-1.3\times10^5$	TrSL3	0.5%	1.2%	3.3 Hz
2-1	Low Re	$7.5\times10^2-3.3\times10^3$	TrSL1/TrSL2	2.5%	6.0%	0.1 Hz
2-2	Medium Re	$2.5\times10^3-1.2\times10^4$	TrSL2	2.5%	6.0%	0.3 Hz
2–3	Medium-high Re	$7.5\times10^3-3.5\times10^4$	TrSL2/TrSL3	2.5%	6.0%	1.0 Hz
2–4	High Re	$2.5\times10^4-1.3\times10^5$	TrSL3	2.5%	6.0%	3.3 Hz
3-1	Low Re	$7.5\times10^2-3.3\times10^3$	TrSL1/TrSL2	5.0%	12.0%	0.1 Hz
3-2	Medium Re	$2.5\times10^3-1.2\times10^4$	TrSL2	5.0%	12.0%	0.3 Hz
3-3	Medium-high Re	$7.5\times10^3-3.5\times10^4$	TrSL2/TrSL3	5.0%	12.0%	1.0 Hz
3–4	High Re	$2.5\times10^4-1.3\times10^5$	TrSL3	5.0%	12.0%	3.3 Hz
4-1	Low Re	$7.5\times10^2-3.3\times10^3$	TrSL1/TrSL2	10.0%	24.0%	0.1 Hz
4-2	Medium Re	$2.5\times10^3-1.2\times10^4$	TrSL2	10.0%	24.0%	0.3 Hz
4-3	Medium-high Re	$7.5\times10^3-3.5\times10^4$	TrSL2/TrSL3	10.0%	24.0%	1.0 Hz
4-4	High Re	$2.5\times10^4-1.3\times10^5$	TrSL3	10.0%	24.0%	3.3 Hz

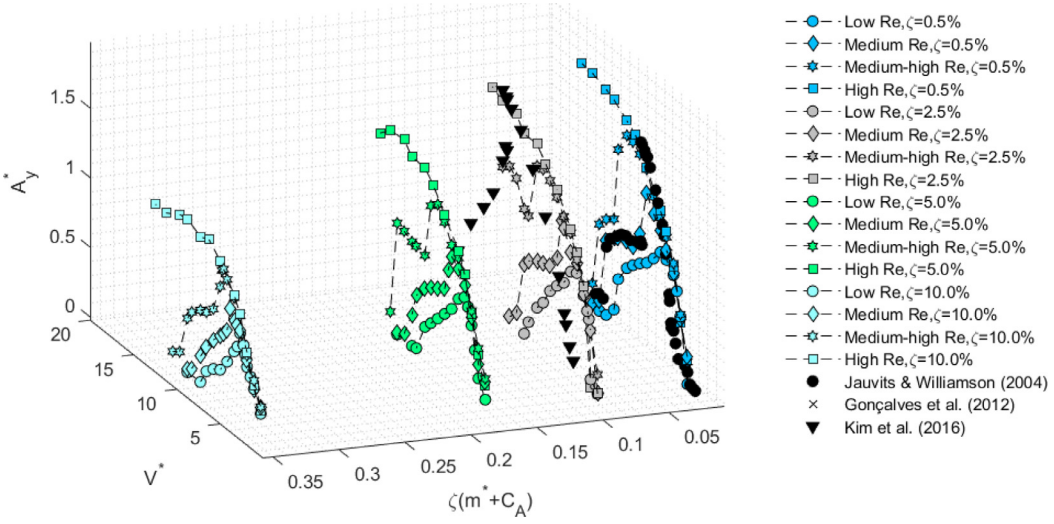


Fig. 5. Plots of cross-flow vibration amplitude versus reduced velocity versus damping-mass parameter. Data from the literature are also shown.

preserved throughout the simulations, avoiding unnecessary and undesired deformations of small mesh elements.

3. Results

The effects of the Reynolds regimes and damping ratios on the structural responses of 2DOF elastically mounted rigid cylinders shall be investigated. Response plots of each DOF reduced amplitude, A_i^*, phase angle, ϕ_i, force coefficients, C_i, and frequency ratio, $f_y/f_{n,w}$, are plotted in the $\zeta(C_A+m^*)$ *vs.* V^* plane. Each system consists on a single combination of TrSL flow regime and damping ratio, and the complete dataset is shown in Table 4. Data from experimental studies from the literature considered for comparison are listed in Table 2. Reduced velocities in the range of $V^*=3$ to 15 are considered. All systems have a mass ratio of $m^*=2.4$. Mechanical responses that are sinusoidal in nature, such as response amplitudes and oscillatory components of force coefficients, are estimated based on the top 20% highest response amplitudes, in steady-state. Scatter plot of a given generic variable, ξ, are plotted in the plane $\zeta(m^*+C_A)$ *vs.* V^*, in the form of $\zeta(m^*+C_A)$ *vs.* V^* *vs.* ξ, and resultant surfaces are interpolated using the Triangulation-based cubic interpolation approach [52].

In Section 3.1 it is focused on the mechanical responses of the cylinder undergoing VIV, characterizing the turbulence and structural damping effects on its dynamics. In Section 3.2, it is focused on the analysis of vortex shedding in the near-wake on VIV.

3.1. Structural responses to different Reynolds regimes flows

In Fig. 5, cross-flow response amplitudes from the complete range of computational analyses performed in this study are shown. Data from the literature is overlaid to our results as well. The highly cited studies from Jauvits and Williamson [5], Kim et al. [36] and Gonçalves et al. [35], with low damping ratios, showed good agreement to our numeric model. These experimental set-ups present slightly higher vibration amplitudes, which is a common relation between numerical and experimental models (see Gsell et al. [26]). Nevertheless, given the fact that our numeric model was able to predict with appreciable agreement results from three relevant experimental studies, it is concluded that other mechanical responses obtained in this study may also well represent those of real models.

In Fig. 6 it is shown the cross-flow response amplitudes for different damping ratios under different Reynolds regimes. Distinct behaviours are observed for the variation of A_y^* by V^* for the Low, Medium, Medium-high and High Reynolds cases. For low Reynolds flows, shown in the top left corner of Fig. 6, the peak response amplitude is found for a mean reduced velocity of $V^*=6$. For higher damping ratios, the maximum amplitude decreases and it is found for lower values of reduced velocity. In the Low Reynolds cases, a smooth decrease is observed from the upper branch to the desynchronized regime. For intermediate Reynolds numbers, in the range of $Re=2.5\times10^3-1.2\times10^4$, in the TrSL2 flow regime, the lower branch becomes present in the responses for high

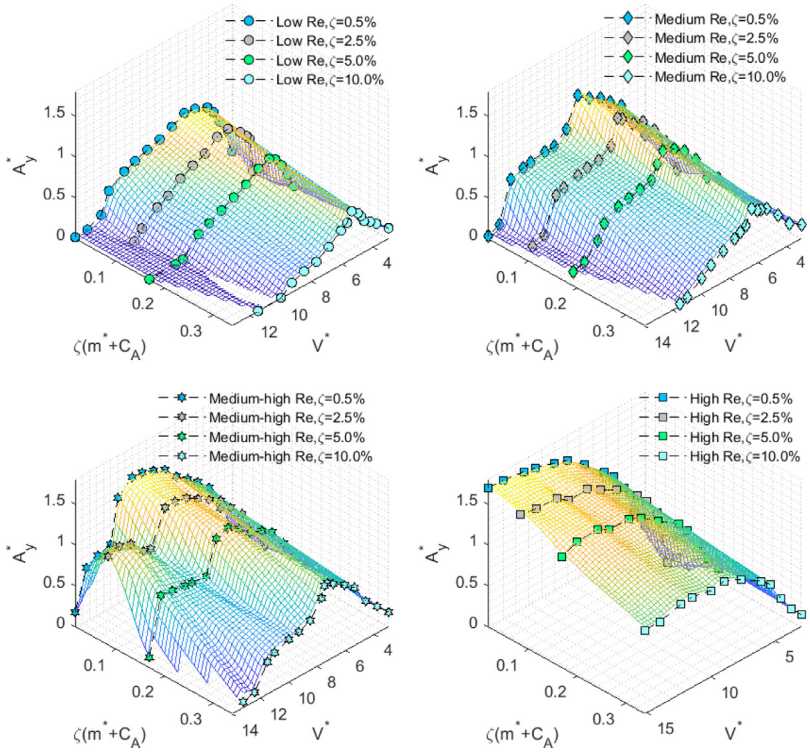


Fig. 6. Cross-flow reduced amplitude, A_y^*, plots as function of $\zeta(m^* + C_A)$ and V^*.

reduced velocities. This behaviour was extensively studied by the past literature (see [1,2,4]). Sequentially, the flows in the end of the TrSL3 regime, the High Reynolds cases, present a continuous increase trend in the range of reduced velocities investigated.

By comparing the responses shown in Fig. 6, it becomes clear that increments in the Reynolds number lead to higher reduced velocities at which the transition from the upper to the lower branch takes place. So far, most of published works agree that the shift of the this transition point is only related to the increment of the upper branch extension, which mechanisms of elongation were discussed by many studies (see [53] and [14]). In our results, on the other hand, we have observed evidences that this may not be necessarily true for the entire TrSL flow regime. Differently from the response changes observed in the TrSL1 and TrSL2 flow regimes by Khalak and Williamson [2], in which the transition point from the upper to the lower branch presents relatively low variation by *Re*, an appreciable increase in the upper branch extension is observed in the TrSL3 regime. Also, from the Medium to the Medium-high Reynolds cases, a considerable increase in the height of the lower branch is observed, mostly for intermediate damping ratio values. These remarks lead us to conclude that in the TrSL3 the response amplitude in the lower branch increases, quickly forming a plateau that joins together with the upper branch, justifying the great elongation observed in the upper branch extension in this regime.

Plots of the stream-wise movement amplitude are shown in Fig. 7. Small displacements are observed for most data points, although some peak movement amplitudes of about $A_x^* = 0.5$ are reached for the Medium-high and $\zeta = 0.5\%$ case, in the TrSL3 regime, which is associated with the upper branch dynamics. By comparing the cross-flow to the stream-wise response amplitudes, in Fig. 6 and Fig. 7, it is possible to notice that A_x^* increases by A_y^* for low damping-mass parameters, $\zeta(m^* + C_A)$, but both DOF become desynchronized when high values of this parameter are implemented. In these cases, the maximum amplitude in the stream-wise direction is found for a constant value of reduced velocity of approximately $V^* = 6$, in despite of the Reynolds regime or cross-flow response amplitude, a completely different trend of that observed for the cross-flow movement dynamics, which has not yet been mentioned in the literature.

The cross-flow force coefficient is investigated and results are shown in Fig. 8. The peak amplitude is found for $V^* = 5$, independently of the Reynolds regime or damping ratio. Curiously, this value of V^* is the reduced velocity in which the resonance of the 1DOF elastically mounted cylinder system occors, that is, $f_y/f_{n,w} = 1$ (see [54]). The cross-flow movement amplitude, shown in Fig. 6, does not vary by the cross-flow force coefficient. Also, the force coefficient presents small variations by the Reynolds number, indicating that C_y is approximately constant in the entire TrSL region for VIV, which is a different result of that classically obtained with fixed rigid cylinders, in which the lift force coefficient value decreases by *Re* [55].

The stream-wise force coefficient, C_x, shown in Fig. 9, is determined by adding the oscillatory component to the mean steady-state drag value, where the former comprises to the 20% highest response amplitudes. For higher values of the *Re* number, the de-synchronization between the cross-flow excitation force and the responses of the cylinder occurs for higher values of V^*, with a smoother transition from $C_x \approx 4$ to $C_y \approx 1$ taking place from $V^* = 6$ to $V^* = 14$, for the High Reynolds case. The initial synchronization point also varies with the TrSL flow regime. For the Low Reynolds case, the initial synchronization occurs approximately at $V^* = 5.3$, for the Medium Reynolds case, at $V^* = 7.7 - 6.1$, for the Medium-high, at $V^* = 8.5 - 7.5$, and for the High Reynolds case, at $V^* = 6.1$. Lower values of $\zeta(m^* + C_A)$ shift the peak value of the stream-wise force coefficient to higher values of reduced velocity, which is a new trend observed in our results.

Another remark observed in our results is in the fact that the C_x curves approximately follow the A_y^* surfaces in the TrSL1, TrSL2 and in the beginning of the TrSL3 regime, but both variables present completely different responses in the end of the TrSL3 regime (see Fig. 6 and Fig. 9). Also, the maximum values of drag coefficient found in the Low, Medium, Medium-high and High Reynolds cases are given by: $C_x = 3.04$, 3.68, 4.34 and 4.06, respectively, meaning that an increase of the total drag coefficient takes place during the TrSL1 and TrSL2 regimes, and it reaches a plateau in the TrSL3 regime. This

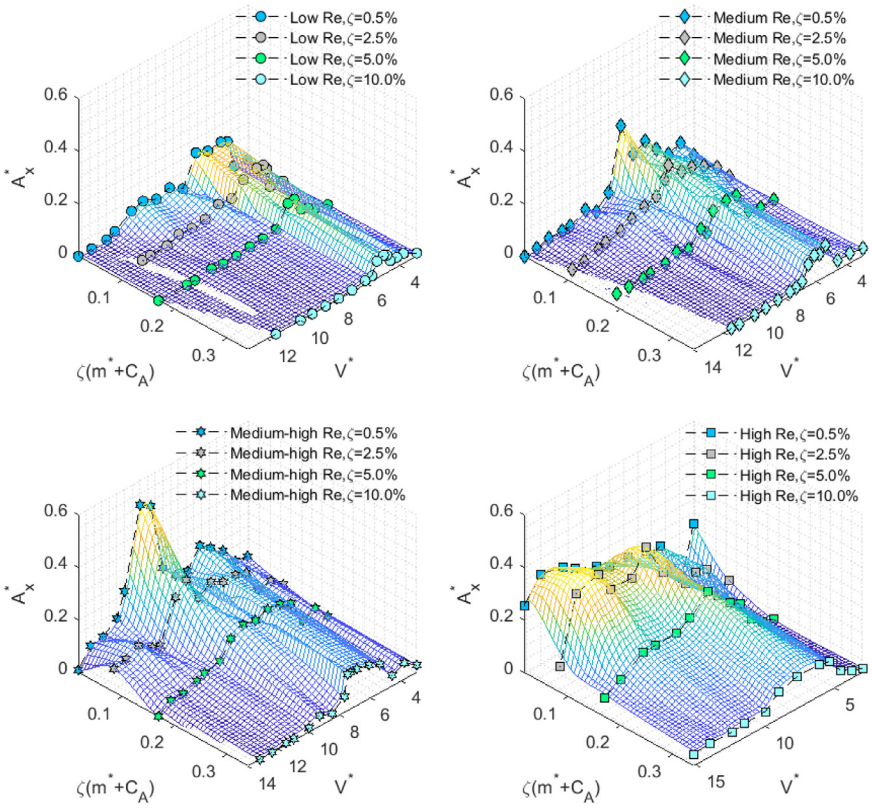


Fig. 7. Stream-wise reduced amplitude, A_x^*, plots as function of $\zeta(m^* + C_A)$ and V^*.

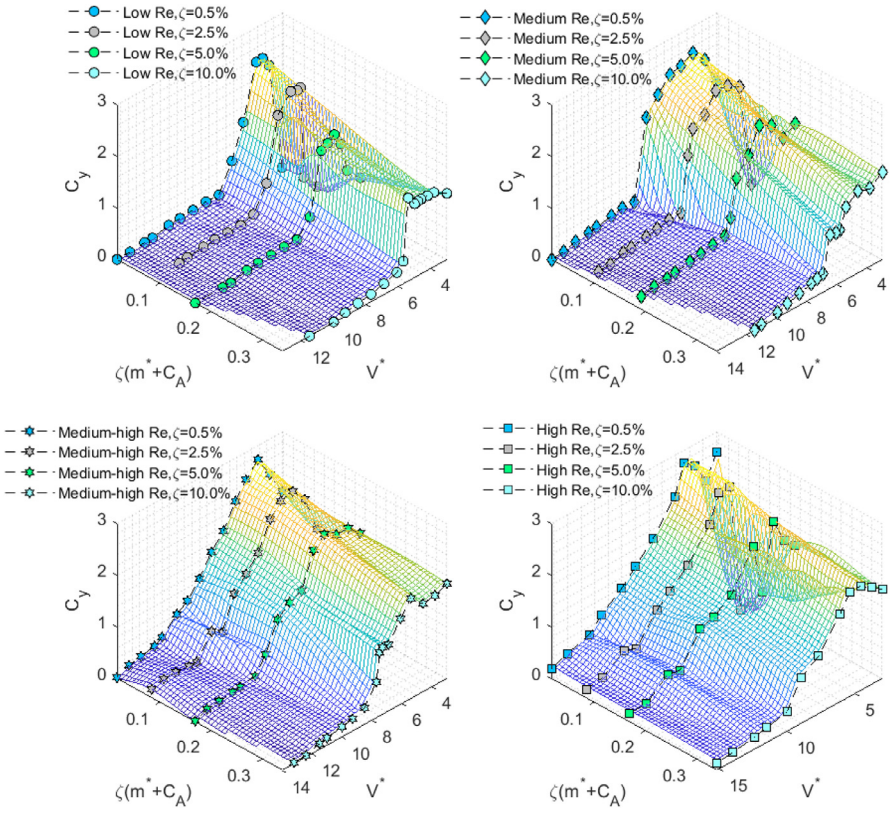


Fig. 8. Cross-flow force coefficient, C_y, plots as function of $\zeta(m^* + C_A)$ and V^*.

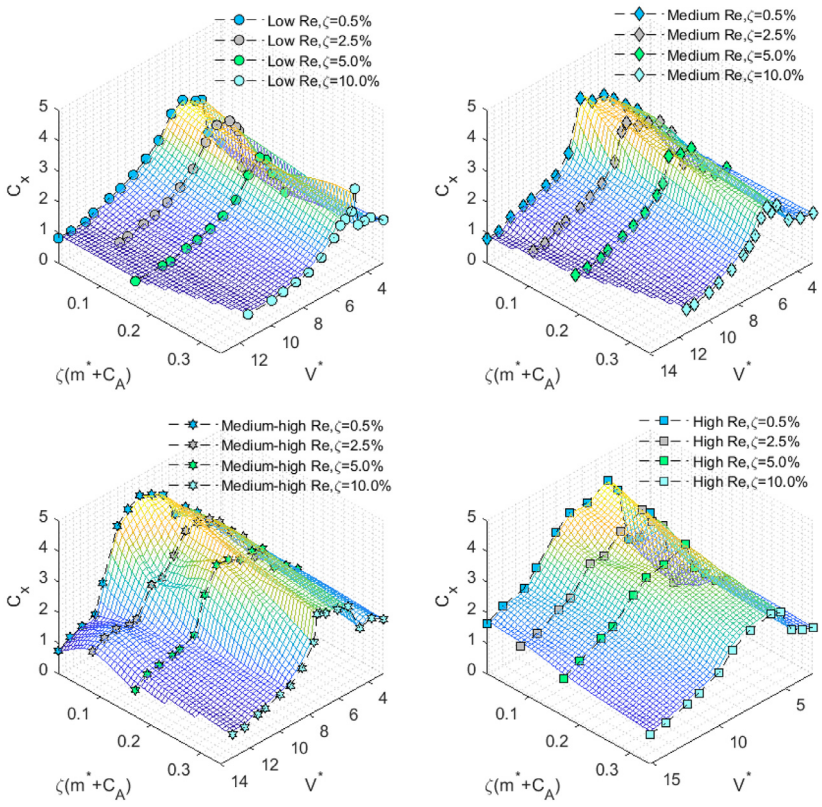


Fig. 9. Stream-wise force coefficient, C_x, plots as function of $\zeta(m^* + C_A)$ and V^*.

trend is similar to that observed for flows around fixed ridid cylinders [31].

For all investigated flow conditions, the frequency ratio, $f_y/f_{n,w}$, presented similar responses, increasing from values of $f_y/f_{n,w} = 0.5$ to approximately $f_y/f_{n,w} = 1.0$, in the TrSL1 and TrSL2 regimes, and to $f_y/f_n = 1.2$, in the TrSL3 regime. These results are shown in Fig. 10. Further, considering a linear function that interpolates data points in the TrSL3 flow regime, we propose that for a expression of the form $f_y/f_{n,w} = A + BV^*$, with constants A and B, the variation of the cross-flow vibration frequency in the TrSL3 flow regime is identified by:

$$\frac{f_y}{f_{n,w}} = A + \frac{St}{\pi} V^* \tag{10}$$

where St is the Strouhal law vibration frequency, given by $St = 0.2$. Some divergences regarding the actual behaviour of the response frequency may be found in the past literature (see [26] and [56], for example). Mostly, there was no agreement on the actual value of the slope of the linear function that interpolates these data points. For instance, while some studies agree that $0.1St$ is a fair approximation for the slope in the TrSL2 regime, such as Assi [53], others found out values close to $0.83St$ [57], also in the TrSL2 regime. Further, Raghavan and Bernitsas [31] have produced experimental results within the highest values of Reynolds numbers that we were able to find, in the TrSL3 regime (see Table 2). Their model presented a value of slope of approximately $B = 0.0512$ for the synchronized regime. In this study, we have found a mean value of $\overline{B} = 0.06364$ for the slopes in the TrSL3 regime, which we have noticed to be appreciably approximated by the expression $B = St/\pi \approx 0.06366$. This interesting result is shown in Fig. 10.

Most of published studies on VIV of elastically mounted rigid cylinders lack of data on response phases, mainly for high Reynolds flows. In this study, we have focused on obtaining the angle between steady-state signals from the cross-flow displacement, y, and excitation force, F_y, which results are shown in Fig. 11. For the Low and Medium Reynolds cases, in the TrSL1 and TrSL2 regimes, the transition from the upper to the lower branch is marked by a phase shift from 0 to 200°. Increments in the damping ratio anticipate these transition.

Another remark observed in our results is related to the fact that, while very low damping ratios allow phase shifts to up to $\phi_y = 200°$, increments in its value decrease these variations to values of up to 140°. For instance, Jauvits and Williamson [5] showed that their system, with very low damping ratio (see Table 2) is only able to shift from $\phi_y = 0$ to 180°, during the upper branch. Furthermore, in our study, increments in the mean Reynolds number did not lead to consistent divergences between response phases, which means that the Reynolds number does not impose a considerable influence on the response phase. On the other hand, this trend is not reproduced in the end of the TrSL3 regime. In this regime, only small phase angle changes of up to $\phi_y = 60°$ are observed, which are also shown in Fig. 11.

3.2. Characteristics of the near-wake flow in the VIV phenomenon

In this section, we shall investigate the patterns of vortex shedding on vortex-induced vibrations of elastically mounted cylinders. Some interesting data points, which are represented in Fig. 12, will be investigate in terms of fluid mechanics. Hereafter, the acronym P, 2S, P + S and 2P are used to describe a pair of vortices, two single vortices, a pair plus a single vortex and two parallel binaries of vortices being shed from the surface of the cylinder by a quarter of oscillatory motion. In this study, it is focused on the near-wake region, since many studies in the past literature were able to satisfactorily classify vortex shedding in the vortex street. All investigated cases are listed in Table 5.

In Fig. 13 it is shown three-dimensional vortex-streets for two different response regimes of the case A1. The three-dimensional illustrations are represented using instantaneous iso-surfaces of the Q-criterion, while the colour-map represents iso-contours of the velocity magnitude. The picture to the left is associated with initial cross-flow movement, in the transient regime, and the picture to the right, to the steady-state high-amplitude dynamics.

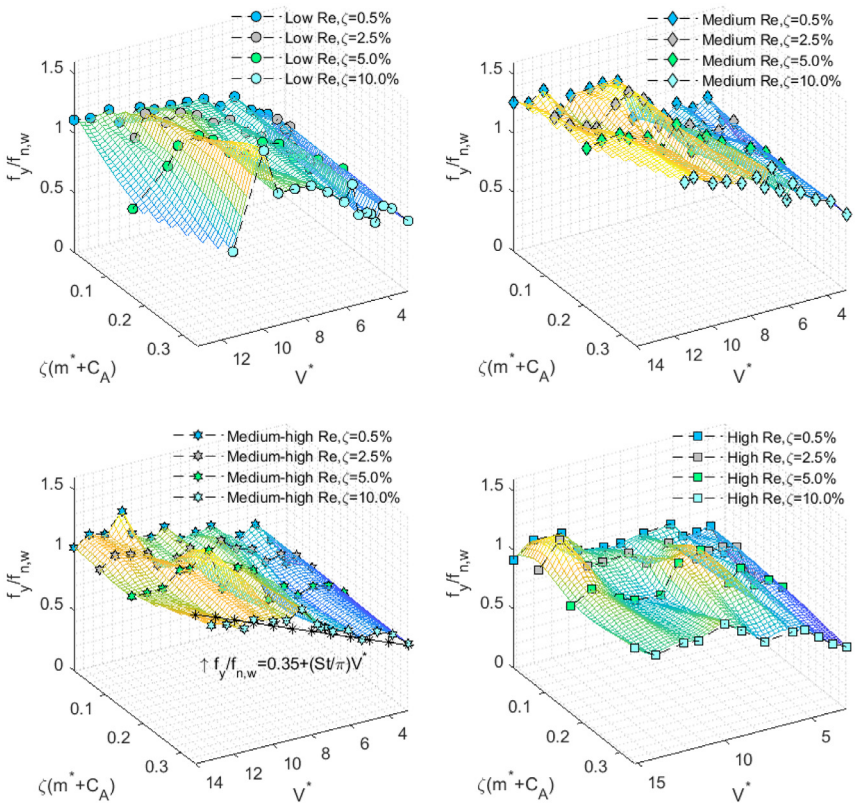


Fig. 10. Cross-flow vibration frequency-ratio, $f_y/f_{n,w}$, plots as function of $\zeta(m^* + C_A)$ and V^*. The proposed interpolation function for the Medium-high Reynolds case, in the TrSL2 flow regime, is also shown.

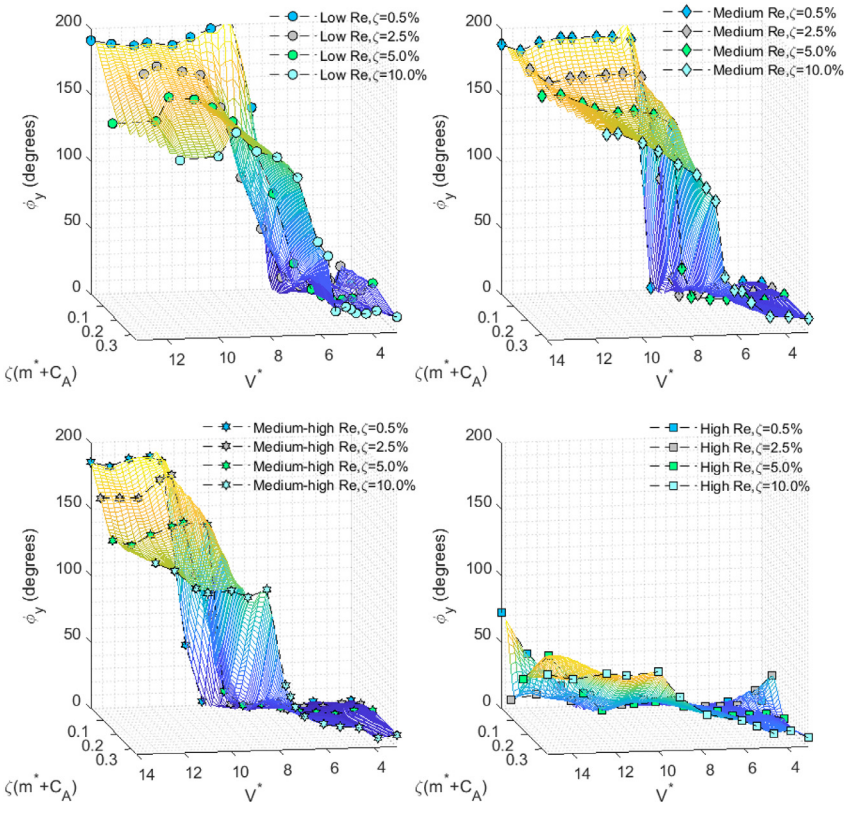


Fig. 11. Phase angle between cross-flow excitation force and response movement, ϕ_y, as function of $\zeta(m^* + C_A)$ and V^*.

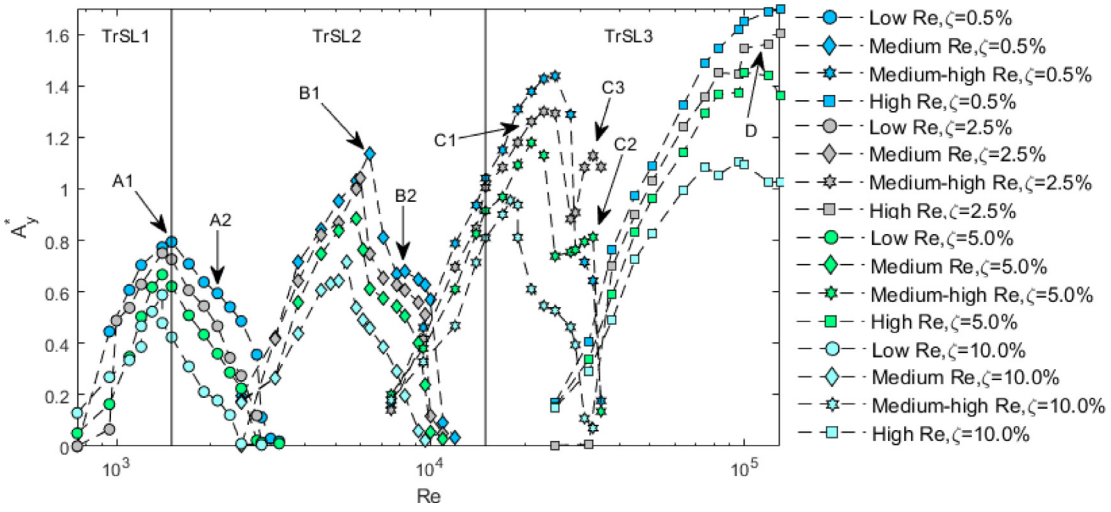


Fig. 12. Cross-flow response amplitude versus Reynolds number for different Reynolds regimes and damping ratios. Highlighted data points are listed in Table 5.

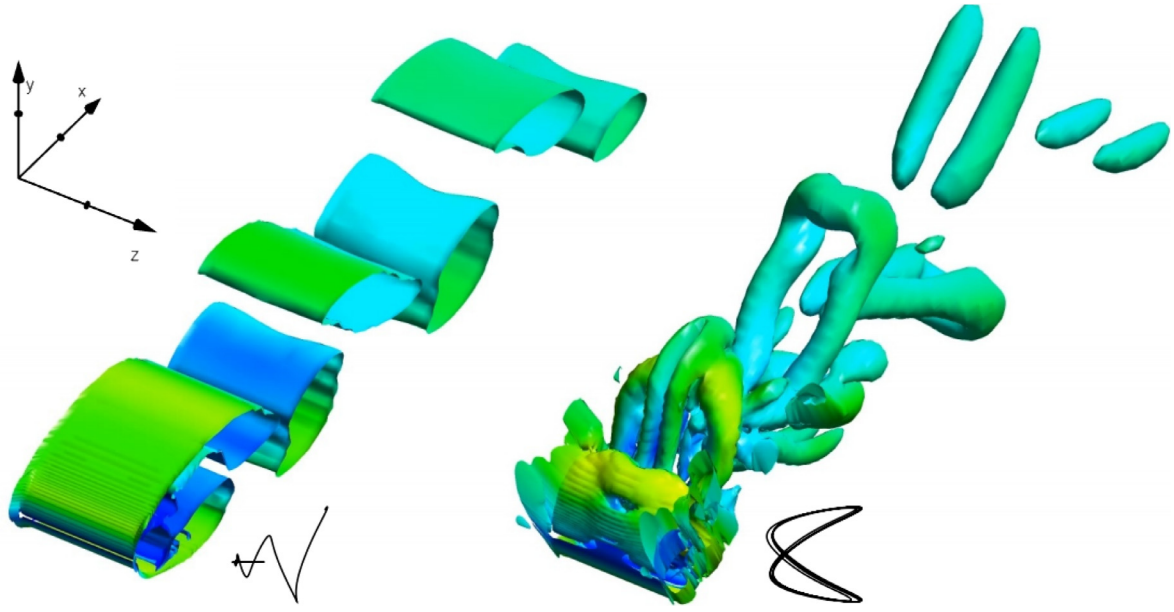


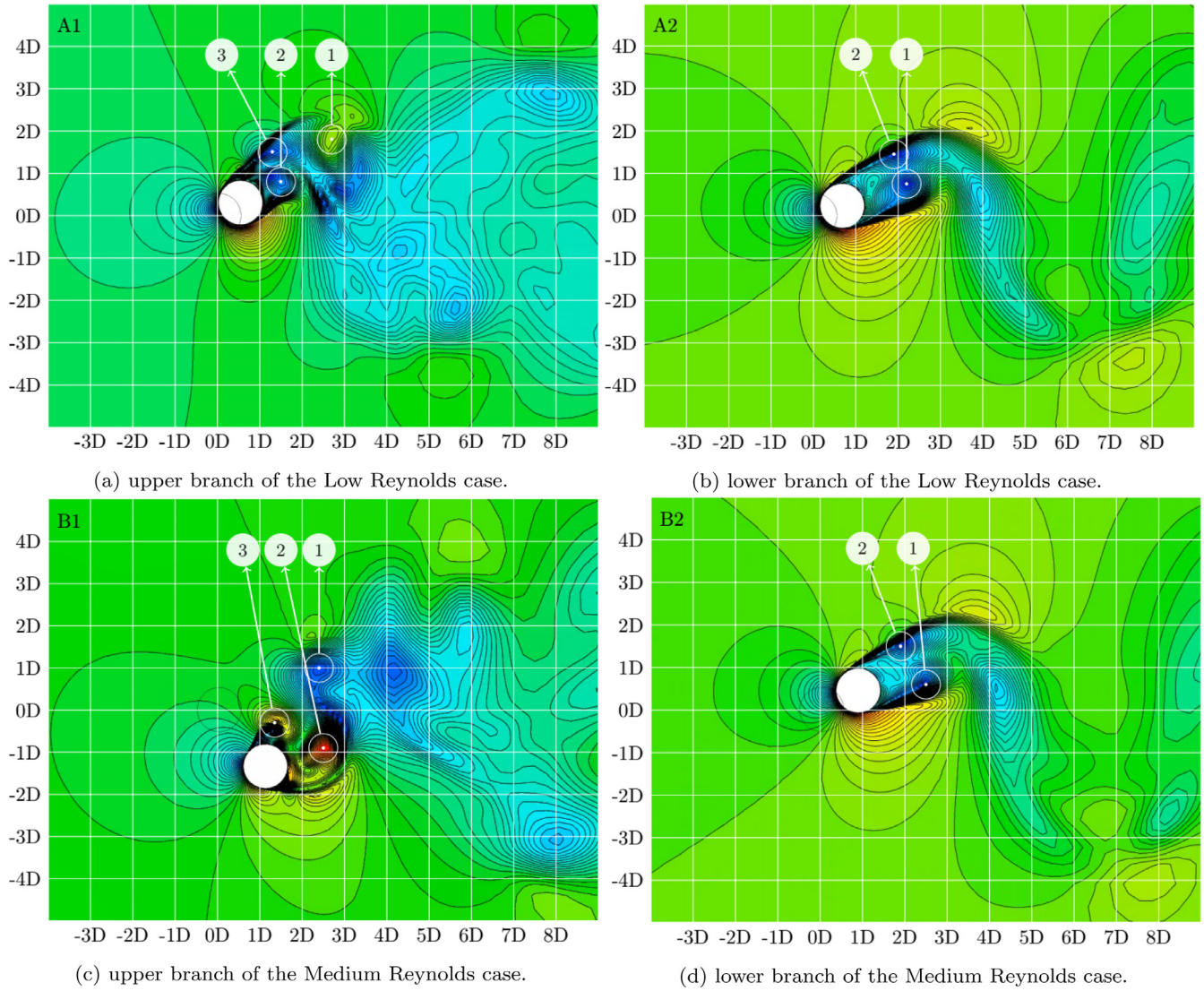
Fig. 13. three-dimensional vortex-street represented by iso-surfaces of the Q-criterion, with normalized strength level of 10^{-2} (please refer to [58]). Colour-map represents iso-contours of the velocity magnitude. (Left) initial low-amplitudes cross-flow movement, (right) steady-state high-amplitudes responses. Both figures are associated with the case A1 in Table 5. The *xy*-trajectories are illustrated next to the cylinders.

Table 5
Characteristics of highlighted data-points in Fig. 12.

Case	A_y^*	Re	$\zeta(m^*+C_a)$	Flow regime
A1	0.794	1.5×10^3	1.2%	TrSL1
A2	0.594	2.1×10^3	1.2%	TrSL2
B1	1.137	6.4×10^3	1.2%	TrSL2
B2	0.677	8.3×10^3	1.2%	TrSL2
C1	1.261	2.1×10^4	6.0%	TrSL3
C2	0.812	3.3×10^4	12.0%	TrSL3
C3	1.129	3.3×10^4	6.0%	TrSL3
D	1.562	1.5×10^5	6.0%	TrSL3

The three-dimensional plots of iso-contours represented in Fig. 13 clearly indicate that the movement amplitude in the *xy*-plane has a strong influence on the transverse-direction flow (*z*-direction), for a constant Reynolds number. The interaction between the stream-lines and the surface of the cylinder makes the shear-layers experience shear flow instability for higher movement amplitudes. For higher values of A_y^*, vortex cores become smaller at first, given the new rotational components inserted in the *xy*-plane, but their scale increase as the flow goes further downstream. For the high-amplitude motion, the flow separation region is small in size, even though stronger vortices are formed with larger movement amplitudes. Note also that the vortex formation region is symmetrical with respect to the *xy*-plane for both illustrated cases, even though the shapes of the vortices considerably change by the movement amplitude.

The results discussed in the previous paragraph are in agreement with those obtained by Gsell et al. [26] and Rosetti and Vaz [27], in which finer three-dimensional structures and larger variation of the flow velocity compared to fixed or small-displacements analyses are observed (see also [59]). The geometry of the shed structures are also in agreement, indicating that the three-dimensionality of the fluid motion is an expected feature on VIV. Thin and long core regions are formed by quarter of cross-flow vibration period and these are shed parallel, as it is also expressed in Figs. 14 and 15. These structures evolves centred-cylindrical vortex formations, further downstream. These



(a) upper branch of the Low Reynolds case.

(b) lower branch of the Low Reynolds case.

(c) upper branch of the Medium Reynolds case.

(d) lower branch of the Medium Reynolds case.

Fig. 14. Contours of velocity magnitude in the symmetry plane of the computational domain. Low and Medium Reynolds cases, listed in Fig. 12, are shown. The sequences of vortex shedding in the near-wake regime are indicated inside the white circles.

described structures are associated with a *xy*-trajectory in an "8-shape" [10] for the cylinder motion, which are described in Fig. 2 and are illustrated in Fig. 13.

Based on the results shown in Fig. 14 and Fig. 15 for the Low, Medium and Medium-high Reynolds cases, the upper branch is associated with the formations of three small vortices by quarter of oscillatory cross-flow motion in the near-wake region. For the dynamics in the upper branch of the Low Reynolds case (A1), which is shown in Fig. 14a, the first vortex sheds from the surface during the end of the upward movement, while vortices 2 and 3 shed during the initial downward movement. These vortices have negative, positive and negative vorticity vectors, respectively.

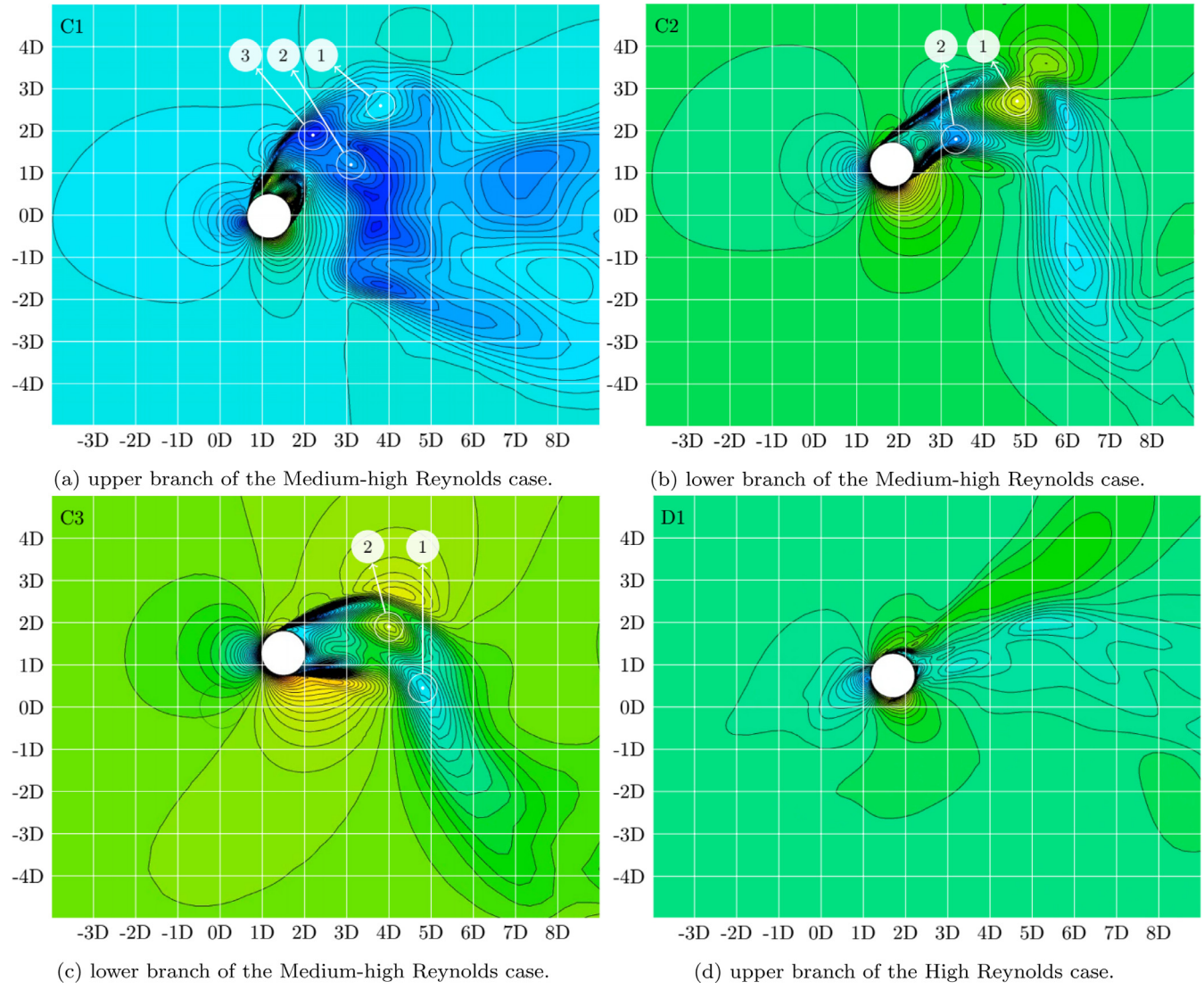
The dynamics in the upper branch of the Medium Reynolds case, shown in Fig. 14c, is associated with three vortices being shed during the downward movement only, with a strong secondary vortex formation during the end of the downward movement. The vorticity of the three structures follow the same trend of the Low Reynolds case. For the Low, Medium and Medium-high Reynolds cases, the length of the vortex-formation region is approximately constant and equal to about 2*D*.

By comparing all velocity magnitude contours of the systems in the lower-branch it is observed two different two-vortex wake patterns: two vortices being shed simultaneously, in the TrSL1 and TrSl2 regime, and two vortices being shed sequentially, in the TrSL3. The former pattern agrees with the results obtained by Morse and Coughran [9], who have also observed the 2P wake structure in a similar Reynolds regime.

In addition, it was mentioned in the previous section that the lower branch appreciably increases its amplitude during the beginning of the TrSL3 regime, specially for intermediate damping ratio values, which are represented by the cases "C2" and "C3" in Fig. 15. The dynamics in the case C2 is marked by a thin vortex formation region, while a remarkably thick structure is observed for the case C3, which is associated with the highest vibration amplitude observed in a lower branch, in our results.

The region of the flow close behind a bluff body plays an important role in the determination of the vortex shedding. During the TrSL regime, mostly for Reynolds numbers from about 1×10^3 to about 5×10^4, the transition to turbulence moves upstream in the shear layers [60], which shrinks the vortex-formation region in size [61]. These regions are clearly identifiable by the region with strong rolling-up shear layers in Figs. 14 and 15. Roshko [62] showed that the frequency of vortex shedding increases if the scale of the formation region is reduced, given that shear layers are brought closer together, facilitating their interaction.

Also, since one should not differentiate between a thick or thin shear layer spread under the action of turbulence diffusion and a shear layer



(a) upper branch of the Medium-high Reynolds case.

(b) lower branch of the Medium-high Reynolds case.

(c) lower branch of the Medium-high Reynolds case.

(d) upper branch of the High Reynolds case.

Fig. 15. Contours of velocity magnitude in the symmetry plane of the computational domain. Medium-high and High Reynolds cases, listed in Fig. 12, are shown. The sequences of vortex shedding in the near-wake regime are indicated inside the white circles.

formed by its rolling up into a concentrated series of shed vortices, as explicitly mentioned by Gerrard [61], it can be induced that stronger and larger vortex formations in the near-wake are associated with thicker vortex-formation regions. These larger vortex, in turn, produce the incremented excitation force that leads to higher response amplitudes in the cross-flow direction observed in the case C3 relatively to the C2, even if identical boundary conditions are implemented in the flows in both cases.

Despite this, there is no reasonable reason for higher damping ratios to produce higher vibration amplitudes. As a matter of fact, the classic theory, which is extensively investigated by Nakamura et al. [63], predicts the inverse tendency for the response of the system based on the reduced damping ratio. In order to solve this issue, the authors have proposed a different view over the response amplitude based on its variation by the alternative reduced damping definition proposed by Vandiver [64]. He has described an interesting alternative description for the reduced damping parameter that leads us to a better prediction of vortex-induced vibrations and allows us to develop a reasonable justification for the observed phenomenon.

For rigid or elastic cylinders with non-time-varying properties, Vandiver [64] defines an alternative form for the reduced damping-parameter, given by:

$$S_u = \frac{4\pi^2 m_t \zeta}{\rho_f D^2 V^{*2}} \frac{L}{L_{in}} \tag{11}$$

where m_t and L_{in} are the total mass per unit length and the power-in total length. This model is an universally applicable alternative of the classic S_G model described by Griffin [65] for uniform and non-uniform flows. A more convenient form of it is given by:

$$S_u = \frac{\pi^3 \zeta}{V^{*2}} \tag{12}$$

considering that $L_{in} = L$, in our model. In Fig. 16, curves of cross-flow reduced amplitudes versus S_u are shown. The desynchronized regime is highlighted in red. The areas in blue indicate the initial excitation regime, being limited by the maximum response amplitude $A^*_{y,\max}$ obtained with each value of ζ. For example, the lighter blue region indicates the desynchronizing regime for all cases with $\zeta = 10\%$. Darker sub-areas indicate the desynchronizing regime of lower ζ cases.

Based on the $A^*_{y,\max}$ *vs.* S_u curves shown in Fig. 16, it is possible to see that the lower branch of the $\zeta = 2.5\%$ Medium-high Reynolds case is located inside the darkest blue area, in which the extension of the lower

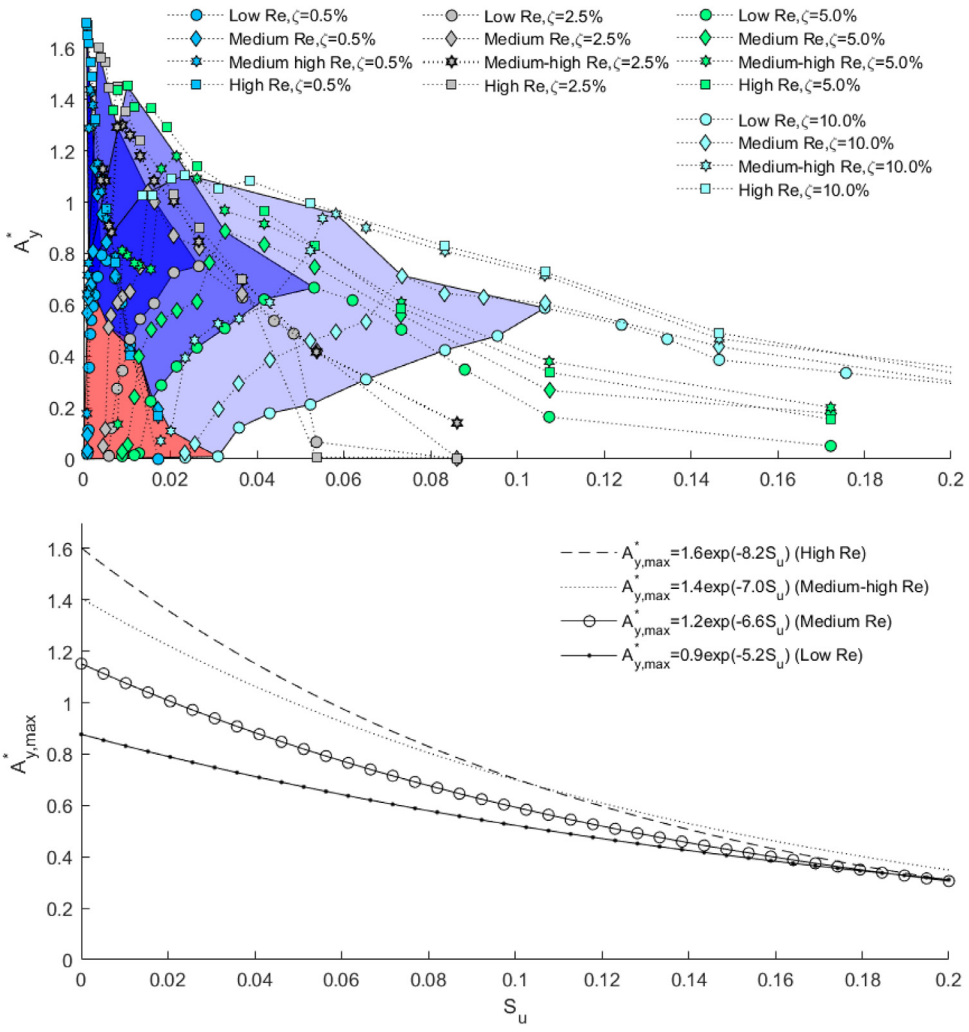


Fig. 16. Curves of cross-flow response amplitude versus reduced damping. The coloured areas indicate the desynchronizing region for each damping ratio tested, being limited by the minimum and maximum values of A_y^* obtained with each value of ζ. The light blue area shows results for $\zeta = 10\%$. Lower values of damping ratio are represented by darker shades of blue. In the second figure, exponential curves that fit the maximum response amplitude $A^*_{y,\max}$ obtained in each regime as a function of S_u are shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

branch becomes small and the slopes greatly increase. As the flow conditions force the dynamics of the system to the far-left region of this map, a strong tendency of cross-flow movement amplitude increase is observed with the decrease of S_u. The relatively high pressure forces anticipate the formation of the upper and super-upper branches and also cause the unexpected increment of the lower branch height in this region, in a condition that the internal forces are overcome by the excitation forces.

In the same figure it is shown exponential curves that fit the maximum response amplitude $A^*_{y,\max}$ obtained in each flow regime. For the Low Reynolds case, the variation of the maximum cross-flow response amplitude by the reduced damping is given by $A^*_{y,\max} = 1.6\exp(-8.2S_u)$. For the Medium, Medium-high and High Reynolds cases, the functions are given by: $A^*_{y,\max} = 1.4\exp(-7.0S_u)$, $A^*_{y,\max} = 1.2\exp(-6.6S_u)$ and $A^*_{y,\max} = 0.9\exp(-5.2S_u)$, respectively. For a general form of $A^*_{y,\max} = A_0\exp(-BS_u)$ for the maximum response amplitude in a given TrSL regime, the term A_0 is well-approximated by the maximum response amplitude obtained with the lowest possible damping ratio, ideally: $A_0 \equiv A^*_{y,\max}|\zeta = 0$. The proportional term inside the exponent, B, is greater for higher Reynolds flows. Based on the trends for these exponential curves, the maximum amplitude becomes more dependent on the internal (structural) forces for higher reduced damping values, while the maximum response amplitudes are limited by vortex forces for low values of S_u, which agrees with classic results.

4. Conclusions

In the present study, vortex-induced vibrations of two degrees-of-freedom elastically mounted rigid cylinders are investigated using the classically damped spring-mass system programmed inside the CFD software. Good agreement between numerical results obtained in this study with experimental models from the literature is obtained with relatively low computational costs. In this study, the responses of the cylinders are visualized as a function of $\zeta(m^* + C_A)$ *vs.* V^*, which allowed the authors to present a better comprehension of responses in a wide range of Reynolds numbers and damping ratios.

The clear tendency of increasing and joining the upper branch together is observed for the lower branch for increasing Reynolds number flows. This mechanism is a reasonable explanation for the great increment in the upper branch extension observed in the TrSL3 flow regime. The universal reduced damping function proposed by Vandiver [64] provides an alternative perspective that allowed us to justify this mechanism based on the trends of the A_y^* *vs.* S_u curves. Besides this, the following concluding remarks obtained in the present study must also be emphasized: i. based on our numeric results, the cross-flow vibration frequency is well represented by the function $f_y/f_{n,w} = A + (St/\pi)V^*$ in the beginning of the TrSL3 regime; ii. based on the A_y^* *vs.* S_u curves, shown in Fig. 16, the desynchronizing region becomes wider for higher values of damping ratio; iii. the maximum response amplitude for a given Reynolds regime may be estimated using $A^*_{y,\max} = A_0\exp(-BS_u)$, where $A_0 \equiv A^*_{y,\max}|\zeta = 0$ and B increases by the Reynolds number.

Further, the vortex-formation region is considerably thicker in the upper branch due to the formation of three vortices. For the TrSL1/TrSL2 cases, one vortex is shed during the upward movement and two others are shed during the downward movement. For the TrSL3 regime, the three structures are shed during the downward (or upward) movement only.

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